Research Papers (ACM) 2 per person

AMAN:
Paper 1: https://dl.acm.org/doi/10.1145/3494580

A Survey on Healthy Food Decision Influences Through Technological Innovations

Paper 2: https://dl.acm.org/doi/10.1145/3686928
From Viral Content to Real-Life Cuisine and Beyond: Examining Teenagers' Interactions with TikTok Food Videos and the Influence on their Food Practices .

PAPER 1 — Marshall et al. (2022): Gaps from Table 4 + Discussion

Gap	What's Missing	Your Opportunity
Social factors ignored	Most food tech focuses on individual preferences (taste, nutrition). Social variables like friends' eating habits, social support, and social influence are barely explored in food tech design	Your project IS about social influence on food choices — directly fills this gap
Grocery shopping context under-explored	The paper explicitly says "more focus is needed on the grocery shopping aspect" — most tech is for personal tracking at home, not in-store	Your project targets the in-store grocery moment specifically
No real user testing	Most systems are evaluated offline with algorithms, not with real users in real contexts	You'll do user research and test with real uni students in real shopping contexts
Family/ household food culture ignored	Almost no research on how household composition (housemates, family) shapes food tech needs	Uni students live in shared houses — housemate influence on food is directly relevant
Environmental factors missing	Distance to stores, food availability, living area characteristics rarely considered	Uni students near St Lucia have specific store access (Woolworths Toowong, ALDI, etc.)

No combined context	No system combines grocery shopping context + social features + personal tracking together	Your design opportunity sits exactly at this intersection

PAPER 2 — Wang et al. (2024): Gaps from Limitations & Future Work

Gap	What's Missing	Your Opportunity
Teens only, not uni students	Study was 13–19 year olds still living at home. Uni students (18–25) have independent food purchasing responsibility	You're studying uni students who make their OWN grocery decisions
Online content only, not in-store action	Study tracks TikTok → food inspiration but doesn't follow through to the actual purchase/cooking moment	Your project bridges the gap: what happens when social media inspiration meets the grocery aisle?
No longitudinal data	Only captured snapshots, not long-term behaviour change	Your prototype could support ongoing behaviour, not just one-time use
Gender imbalance	Mostly female participants	Recruit diverse uni students
Individual use only	TikTok food content is consumed alone — no collaborative or shared features explored	Your design can add social accountability, shared lists, group goals
No connection to physical context	Social media food content exists in a vacuum — not connected to where you are, what's available, or what you can afford	Context-aware features (location, store inventory, budget) bridge this gap

YOUR COMBINED DESIGN OPPORTUNITY (use this language in your slides):

Existing research shows that food choice technology focuses on individual preferences and ignores social influence, and that social media food content inspires people but fails to translate into real purchasing behaviour. No existing work combines social features with context-aware support at the point of grocery

purchase for university students. Your project sits in this gap — exploring how social and contextual computing can support healthier food choices when uni students are physically shopping for groceries.

Anujitha:

1. Social influence has mainly been studied in online grocery environments

Di Cosola III and Neff (2022) investigated in-group and out-group social-comparison nudges for healthier food choices. Their experiments were conducted in an online grocery-shopping context, including social comparisons at checkout. The study suggests that social comparison can influence food decisions, but it does not examine how these mechanisms work when people are physically walking through a supermarket and making product decisions.

Research gap: There is an opportunity to investigate whether social-comparison mechanisms can support healthier choices during in-person grocery shopping, particularly when users are actively considering and selecting products.

2. Existing in-store food technologies focus primarily on the individual

Strecker et al. (2024) developed ShoppingCoach, a diminished-reality system that identifies supermarket products and visually reduces the prominence of products that deviate from dietary recommendations. Their study demonstrated that technology can support healthier decisions in an offline supermarket scenario, with 12 participants and improvements in compliance with dietary recommendations and decision time.

However, ShoppingCoach primarily considers the individual user's nutritional requirements and does not incorporate social influence, peer preferences or social-media-derived information.

Research gap: Existing in-store technologies provide an opportunity to investigate how individual nutritional support could be combined with social information, such as peer preferences, social norms or social-media influence.

3. Social-media food content has been linked to offline behaviour, but not specifically to grocery purchasing

Wang, Sher and Chung (2024) examined how teenagers interact with TikTok food videos and how these interactions influence their food practices both online and offline. Through 15 semi-structured interviews with teenagers aged 13–19, the study identified actions including planning, executing food-related activities, sharing content with friends and developing longer-term changes in food-related attitudes and habits.

However, the study focuses on teenagers' food practices rather than university students' independent grocery-shopping decisions. It also does not investigate the transition from social-media inspiration to the specific moment of purchasing a product in a physical supermarket.

Research gap: There is an opportunity to examine how food-related social-media exposure translates into actual grocery-purchasing decisions among university students.

4. The connection between social media and physical shopping context remains underexplored

Wang et al. (2024) demonstrate that TikTok food content can influence activities beyond the social-media platform, including planning and offline food practices. However, the study's research setting remains centred on teenagers' experiences with TikTok and the subsequent food practices inspired by that content.

Similarly, ShoppingCoach demonstrates the potential for contextual technology inside a physical supermarket, but its intervention is based on nutritional information rather than social-media or social-computing mechanisms.

Research gap: There is a clear opportunity at the intersection of these two areas:

Social-media influence + contextual mobile technology + physical grocery shopping

Our project can explore how information originating from or inspired by social interactions could be presented at the point where a student is actually making a food-choice decision.

5. Limited research specifically focuses on university students

Wang et al. (2024) studied teenagers aged 13–19, while Di Cosola III and Neff (2022) studied adults in online grocery experiments. ShoppingCoach, meanwhile, demonstrates an in-person supermarket application but uses a small sample of 12 participants.

This leaves an opportunity to investigate the specific context of younger university students, who may have greater responsibility for their own grocery

purchasing than teenagers and may experience different constraints around budget, convenience, time and social influence.

Research gap: More research is needed on how younger university students respond to social and contextual interventions during their own grocery-shopping decisions.

Referred Papers,

Di Cosola III, B. M., & Neff, G. (2022). Nudging behavior change: using in-group and out-group social comparisons to encourage healthier choices. 2022 CHI Conference on Human Factors in Computing Systems (CHI 2022).

Strecker, J., Wu, J., Bektaş, K., Vaslin, C., & Mayer, S. (2024). ShoppingCoach: Using Diminished Reality to Prevent Unhealthy Food Choices in an Offline Supermarket Scenario. Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), 1–8

https://dl.acm.org/doi/10.1145/3613905.3650795

Kenan

Nudging Behavior Change: In-Group and Out-Group Social Comparisons

- Study tested whether showing shoppers what "people like you" (in group) versus "people not like you" (out group) typically choose changes their own food decisions, applies social comparison theory to a digital grocery shopping context
- Found that social comparisons could increase the nudge people make to change compared to non-social comparisons
- non-social nudges (like simply suggesting swaps) "had mixed results and did not lead [to] a significant reduction in overall calories"
- earlier research found that nudge-style food labels in a "university eatery setting" produced only "small but significant increases" in healthier purchases — useful evidence that student populations respond to non-social nudges
- **Gaps from the article**
    - entirely tested in an online grocery context, not in-person.

        - that out-group comparisons may be just as effective as in-group comparisons in encouraging socially responsible decisions and should be further explored
        - authors state they "test if adding social dimensions to existing nudging interventions can improve nudge uptake" - **underused nudge type in HCI**
        - they specifically tested out-group comparisons (not just the standard "people like you" framing)
        - deliberately tested at **point of checkout**, arguing less is known about nudging at this specific moment compared to nudging during browsing – useful if we want to focus on a specific moment/decision rather than whole shopping
        - social comparison nudges are **transparent** to the user, unlike nudges that manipulate behaviour "outside of their awareness" - for ethical design

ShoppingCoach: Using Diminished Reality to Prevent Unhealthy Food Choices in an Offline Supermarket Scenario

- ShoppingCoach is a prototype that uses diminished reality (removing/blurring things from the shopper's view). It reduces the visual presence of grocery products it identifies as unhealthy based on the products nutrition info and individual user's dietary needs.
- Tested on uni students in an in person supermarket scenario.
    - participants rated their confidence in making healthy purchasing decisions at supermarkets around 3.42/5 — moderate confidence
- **Gaps from the article**
    - No social dimension at all – System only to the individual shopper
    - controlled experiment, not a real-world longitudinal deployment
    - mostly Business/Finance/Economics students tested
    - Author plans to improve system usability and implement an additional AR mode. Roadmap focused on individual/technical side no social direction.

James

Strategies to Promote Healthy Eating Among University Students: A Qualitative Study Using the Nominal Group Technique

- Main problems were money, high prices of healthy food, lack of time, poor food knowledge and social influence.
- This is especially relevant to first-year students, because they may have less experience with food shopping, prices and making their own purchasing decisions.
- One of the biggest social barriers was negative influence from social networks, and this was actually the top objective barrier for freshmen.
- The study also found that students felt Instagram and Facebook could negatively influence their food choices.
- This supports our idea because it shows that social media/social networks are already linked to students' food choices, but it doesn't really look at what happens when they're actually grocery shopping.
- The paper suggests future research should create and test **tailored interventions**, which could be where our social/contextual computing idea comes in.

Key points to use for our topic

- University = major change in students' eating habits.
- First-year students may have less shopping experience.
- Social networks can influence healthy eating.
- Social media was seen as having a negative influence on food choices.
- Price, time and knowledge could also affect how students respond to social-media influences.
- Could use social technology as a way of supporting healthier choices rather than just viewing social media as a problem.

Healthy Eating, Identities, and Design Opportunities in Social Media Spaces

- This paper is more focused on social media and how it affects food and health behaviour.
- It found that teenagers use TikTok food content for food inspiration, social connection and identity, and that this content can affect offline behaviour and longer-term food habits.

- A big thing from this paper is that algorithms are helping decide what food content people see, so people can repeatedly be exposed to certain foods, trends or ideas.
- Social media isn't always negative either. It can give people support, community and motivation around food and health.
- The paper is also directly related to HCI/social computing, because it looks at how social-media features could actually be designed to support healthier eating and reflection.
- The next stage of the research is testing Instagram Story features to see whether social prompts can encourage people to reflect on their food habits.

Key points to use for our topic

- Social media can influence **food behaviour outside of social media**.
- Food videos/posts can give people ideas about what to eat or buy.
- Algorithms can influence what food content people are exposed to.
- Social media can be both **positive and negative**.
- Social features could potentially be used to encourage healthier behaviour.
- This gives us the **social computing/HCI side** of our topic.

Joseph

Food Literacy while Shopping: Motivating Informed Food Purchasing Behaviour with a Situated Gameful App

- Gamified food literacy and diet tracking to improve people's nutrition knowledge, health beliefs, and in-store shopping behaviour.
- "*After making purchases high in nutrients, such as soy sauce that is high in sodium or cheese that is high in saturated fats, many of our participants decided to limit their intake to better match daily recommendations. We take this feedback as indicative that participants were internalizing lessons learned in the store, and practising Food Literacy skills later at home*"

- *"Another limitation was [didn't] address the issue of budgeting. Food insecurity, defined as uncertain or insufficient access to food because of financial limitations, is a widely identified issue for students in the nutrition literature"*
- *"However, both apps encourage planning as a means of maintaining a budget, and present advice to the user to help plan based on cost."*
- Food literacy metrics determined from Nutrition: concepts and controversies (4th edition). Static data.
- Ideas for our prototype: can follow the same idea but include budgeting constraint and use dynamic data. I.e have users track their own body type, fitness, and diet and compile a repository for an AI model to provide recommendations in accordance with health heuristics (see following paper).

Design and Evaluation of Technologies for Informed Food Choices

- Proposes food literacy heuristics. Table 7 in paper.
- Top 3 ranked by nutrition experts (or so they claimed)
    a. **[H2] Interpret Nutrition Content** Help customers interpret a product's nutrition content using symbols, stamps, or colors (e.g., Traffic light colors, Guiding Stars, "High in" symbols)
    b. **[H20] Support budgeting** and place emphasis on healthy items. (e.g., highlight healthy items on sale; have a "Sort by" feature combining lower price and more nutritious items).
    c. Next three are tie for 3rd place: **[H4]** Sort by Nutrition Values, **[H5]** Filter by Nutrition Content, **[H11]** Follow Food Guides
- *"we expected that the retail food industry may be the largest barrier to adoption; many in the food industry already spend considerable resources promoting less healthy yet profitable foods [59] and are known to mislead consumers to increase profits [88]. However, the nutrition literature shows that the promotion of healthier foods can increase profit [19, 67], and our study instead pointed to HCI experts' misconceptions and differences in perspective as a barrier to adoption in practice. Thus, there is a need to advocate within the HCI community for a stronger consideration of individual and public health considerations when designing technology, and for cross-pollination between the HCI and nutrition communities of practice."*

- What this means? Idk. Maybe we for our prototype we can mention potential sponsorship or sumshi from corpos and do integration with their online stores like coles online and import nutritional info and shit.

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Minimum 2 Interviews per person (Total 10 interviews):

Interview questions review:

2. **"Walk me through your last grocery shop from start to finish — how did you decide what to get, where to go, and what actually ended up in your basket?"** (covers Q3, Q4, and opens the whole experience)
3. **"Tell me about a time your grocery shopping habits changed — what caused the shift and how has it stuck or not stuck?"** (covers Q1, evokes a story)
4. **"Describe what it's like when you shop with someone else versus alone — can you think of a specific trip, where having someone there changed what you bought?"** (covers Q2, Q7, Q8)
5. **"Think about the last time you saw food content on social media or from a friend — did it actually change what you cooked or bought? Walk me through what happened.** (covers Q5 without leading)
6. **"How does your living situation shape what you eat week to week — tell me about a typical week of meals and where the food comes from."** (covers Q9, reveals values around health, budget, convenience)

ACTION ITEMS:

- Team Name
- Setup GitHub repo and add course coordinators
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